

3.1 General properties of waves — 5054 Paper 2

Second audit. 2015–. Only 3.1 subquestions (energy transfer, terms, $v = f\lambda$, transverse/longitudinal, reflection, refraction due to speed, diffraction, ripple tank).

2015

May/June 5054/21

Question 4 • included (a), (b), (c)

- (a) State the unit in which pressure is measured.

.....[1]

- (b) Fig. 3.2 shows the axes for a graph of pressure against volume for the air in the syringe.

One point is plotted on the graph at pressure of P_0 and volume V_0 .

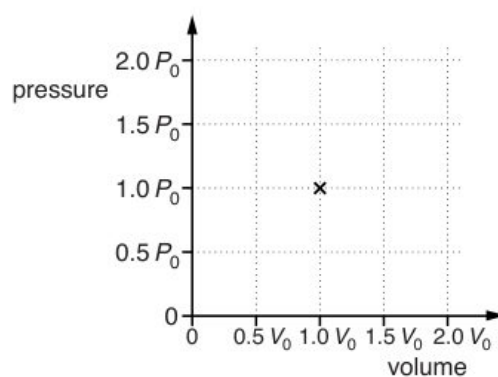


Fig. 3.2

The temperature of the air is kept constant.

On Fig. 3.2,

(c) More metal weights are placed on top of the syringe.

Explain how the molecules of air inside the syringe are able to support more metal weights.

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.....[3]

Page 2	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	21
1	(a) (i) 60 m		B1
	(ii) 12 s		B1
	(b) (i) straight line from origin to 200 m at 40 s		B1
	any line straight or curved from (40,200) to (60,500)		B1
	(ii) $s = d/t$ or $500/60$		C1
	8.3 m/s		A1
2	(a) (i) force moves through a distance (in same direction)		B1
	(ii) chemical (potential) energy		B1
	(b) (i) 480 Nm		B1
	(ii) attempt to apply moments with two forces and distances		C1
	400 N		A1
3	(a) Pa or N/m ² or cm of mercury or atmosphere(s)		B1
	(b) correct points plotted at (0.5V ₀ , 2P ₀) and (2V ₀ , 0.5P ₀)		B1
	curve through points of decreasing gradient		B1
	(c) molecules hit sides/piston		B1
	more molecules hit per second/hit more frequently		B1
	molecular impacts create large(r) force (upwards on piston)		B1
4	(a) oscillate/vibrate stated or described		B1
	transverse movement described		B1
	(b) 0.40 m		B1
	(c) (i) $v = f\lambda$ or $(f =) v/\lambda$ or $2/(b)$		C1
	5.0 Hz		A1
	(ii) clear attempt to draw wave moved along 0.20 m to right		B1
5	(a) $\sin i/\sin r$ or $\sin 50/\sin 30$		C1
	1.5(321)		A1

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May/June 5054/22

Question 9 · included (a), (b), (c), (d)

(a) Explain how rubbing causes the balloon to become negatively charged.

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.....
..... [2]

(b) Explain why the hair is pulled towards the balloon.

.....
.....
.....
..... [2]

(c) Explain why it is important that the balloon is made from an electrical insulator.

.....
..... [1]

(d) State one example where static electricity is useful.

.....
..... [1]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2015	5054	22

- 7 (a) to provide a complete circuit (with live) B1
or to pass current back to mains
or provide a return path for the current
- (b) current/charge/electrons flow to earth/earth wire/ground (when live touches case) B1
fuse melts/blows **and** disconnects circuit/cuts live/stops current B1
- (c) doubly insulated B1
or case/body made of plastic/insulator/not made of metal
or user cannot touch metal
- (d) (circuit breaker) B1
 - turns off/acts fast(er)
 - can be reset
 - easy to see it has tripped/switched
 - can detect small difference between live and neutral currents / small (leakage) current to earth
- 8 (a) left column both 1 B1
right column 0 **and** 1 B1
- (b) (at least one of the atoms) contain same number of electrons and protons B1
or have 1 electron and 1 proton
charge on electron and proton opposite B1
or electron negative **and** proton positive
or charge on electron neutralises/cancels/balances proton charge
neutrons have no charge B1
- 9 (a) number of waves (that pass a point) M1
or number of oscillations (passing a point)
in unit time **or** per second **or** in 1 second A1
- (b) (i) 1.5 cm B1
- (ii) $(v =)f\lambda$ **or** 5×1.5 seen C1
or $(s=)d/t$ **and** $f = 1/t$
7.5 cm/s A1
- (c) (i) wavelength decreases B1
travels a shorter distance in the same time B1
or frequency stays the same (and $v = f\lambda$)

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Question 7 · included (a), (b)

not (c) ultrasound use

(a) Define *boiling point*.

.....
..... [1]

(b) (i) Describe the molecular structure of a liquid.

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.....
.....
..... [2]

Mark scheme — 5054/21 Q7

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2015	5054	21
5	(a) temperature at which a liquid becomes a gas	B1	
	(b) (i) molecules close together/touching or closer than in gas randomly arranged or irregular structure	B1 B1	
	(ii) to separate/increase the distance between molecules work done against (intermolecular) forces or supply p.e. or break bonds	B1 B1	[5]
6	(a) distance from (optical) centre to focal point (principal focus)	B1	
	(b) (i) both Fs correctly positioned at ± 1 mm	B1	
	(ii) two of: paraxial ray to lens through focal point to image ray through optical centre ray through focal point and then paraxial to image (ign. arrows) X at crossing point of rays	M2 A1	
	(iii) 3.4–3.8 cm	B1	[6]
7	(a) at compression: molecules closer together or pressure higher or vice versa for rarefaction	B1	
	(b) (i) $v = f\lambda$ or in words	B1	
	(ii) larger and because the frequency is lower	B1	
	(c) states one use basic idea	(e.g. prenatal scanning) (e.g. ultrasound reflects off foetus)	B1 B1 [5]
8	(a) (i) $(I =)V/R$ or $12/(6000 + 2000)$ or $12/8000$ or $12/2000$ or $12/6000$ or in (ii) $(V =)IR$ or 0.0015×6000 1.5 mA	C1 A1	
	(ii) 9.0 V	B1	
	(b) (reading/it) increases resistance of LDR falls	B1 B1	
	(c) light meter/sensor or automatic light switch or something sensible	B1	[6]
			[45]

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Question 5 • included (a), (b)

- (a) The student plots a graph of the extension of the spring against the mass hanging on the spring. Fig. 2.2 shows the student's graph.

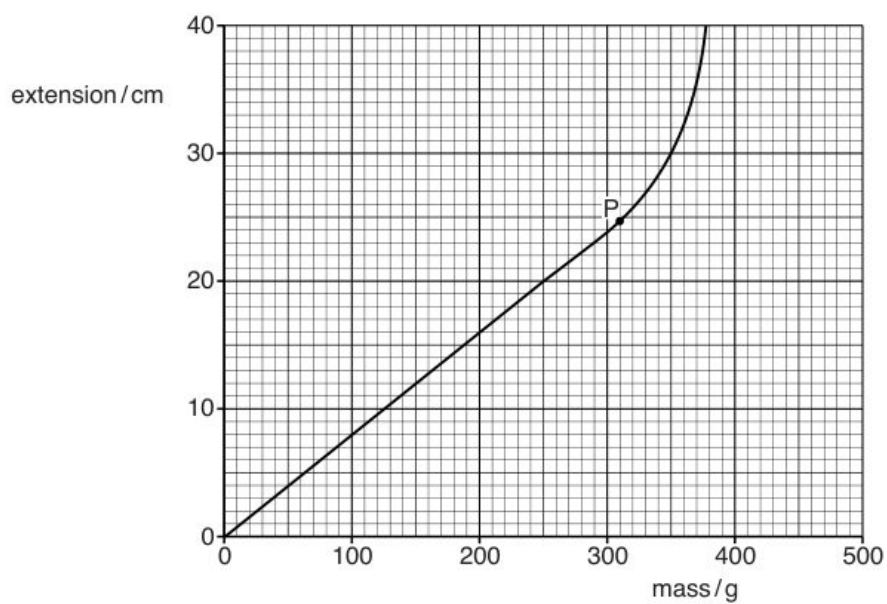


Fig. 2.2

(b) An identical spring is used with the original spring, as shown in Fig. 2.3.

Together they support the mass X.

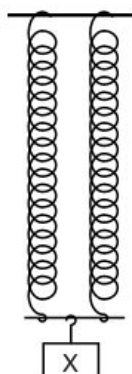


Fig. 2.3 (not to scale)

State and explain how the extension in Fig. 2.3 compares with the extension in Fig. 2.1.

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.....

.....

.....[2]

Mark scheme — 5054/21 Q5

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	21

- (b) (i) 1 difference between smallest and largest temperature
or from 0 to 100 °C B1
- (i) 2 small/moderate distance between (thermometer) marks B1
or for a given temperature change there is a small expansion of liquid / distance
(along scale) / change in thermometric property
or cannot measure small temperature difference / change
- (ii) • use liquid that expands more B1
• smaller bore / thinner tube
• more mercury (in bulb) or use larger bulb
- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time / period for one wave(length) / cycle constant B1
or each oscillation / cycle takes one second
- 6 (a) angle of incidence B1
smallest angle for light to be totally internally reflected B1
or largest angle (of incidence) for ray to be refracted / emerge
or when light emerges along surface
or when angle of refraction is 90°
- (b) (i) $n = 1 / \sin C$ algebraic or numerical C1
2.5 or 2.46 or 2.458(59) A1
- (ii) *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface B1
right hand diagram reflected horizontal ray (by eye) B1
right hand diagram rest of ray completely correct to emerge into air at top face without refraction (by eye) B1
- 7 (a) (current in coil) creates magnetic field C1
or current is at right angles to magnetic field (of permanent / cylindrical magnets)
- (b) into and out of magnet B1
or left and right
or backwards and forwards
current is one way then reverses (so reverses force) B1

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Question 7 • included (c)

only (c)

- (c) The loudspeaker produces sound of frequency 500Hz. The speed of sound is 320m/s. Calculate the wavelength of the sound produced.

wavelength =[2]

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5054/21/M/J/16

Mark scheme — 5054/21 Q7

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	21

- (b) (i) 1 difference between smallest and largest temperature
or from 0 to 100 °C B1
- (i) 2 small/moderate distance between (thermometer) marks B1
or for a given temperature change there is a small expansion of liquid / distance
(along scale) / change in thermometric property
or cannot measure small temperature difference / change
- (ii) • use liquid that expands more B1
• smaller bore / thinner tube
• more mercury (in bulb) or use larger bulb
- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
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- (ii) time / period for one wave(length) / cycle constant B1
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- (b) into and out of magnet B1
or left and right
or backwards and forwards
current is one way then reverses (so reverses force) B1

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Question 10 • included (b)

only CRO amplitude/frequency

(b) Calculate the deceleration of the car at $t = 3.0\text{ s}$.

deceleration =[2]

Mark scheme — 5054/22 Q10

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge O Level – May/June 2016	5054	22

(c)	current is (directly) proportional to voltage or voltage/current is a constant law holds for constant physical conditions/ constant temperature/constant pressure/for metals	B1 B1
(d) (i)	(directly) proportional or $(R) \propto 1$	B1
(ii)	inversely proportional or $(R) \propto 1/A$	B1
(e)	1 st band orange 2 nd and 3 rd bands both black	B1 B1
10 (a) (i)	B – anode D – filament or heater E and F–Y plates or X plates in either order	B1 B1 B1
(ii) 1	attract electrons or gives electrons speed/K.E.	B1
2	heats up cathode or gives electrons energy to escape (metal/cathode) or causes/allows thermionic emission	B1
(iii)	kinetic energy to light or electrical energy to light	B1
(iv)	voltage/charge is applied to the X-plates/vertical plates or turn on time base (steadily) increasing voltage/charge applied to plate(s) or saw tooth voltage applied or electrons attracted/repelled by plate(s) or by the electric field between them	B1 B1
(b) (i) 1	1(.0)V	B1
2	one wave 1.3–1.4 squares or 3 waves in 4 squares 2.6–2.8 ms	C1 A1
3	(f =) 1/T numerical or algebraic 345–400 Hz	C1 A1
(ii)	smaller amplitude shown larger period shown	B1 B1

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Question 5 • included (a), (b)

not (c) production/vacuum

- (a) Explain what is meant by the term *critical angle*.

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.....[2]

- (b) The lamp sends light towards the surface of the pool.

Fig. 4.1 shows three rays of light that are at 30° , 60° and 90° to the horizontal.

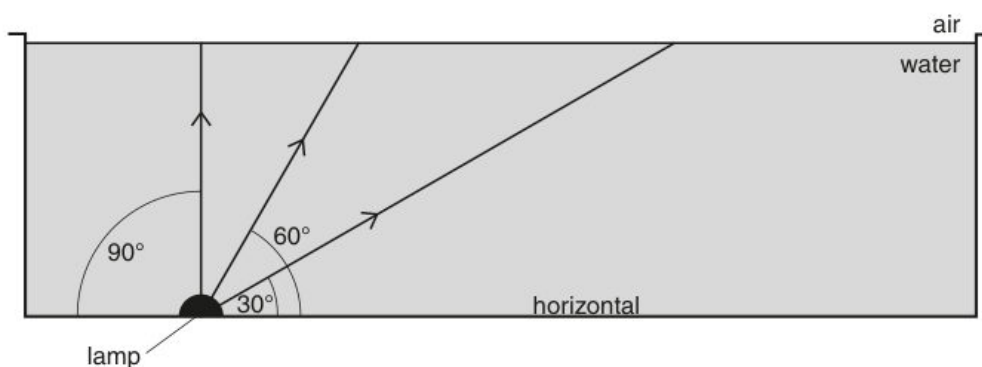


Fig. 4.1

On Fig. 4.1, draw the path taken by each of the three rays after they strike the surface of the water. [3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge O Level – October/November 2016	5054	21
5	(a) number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time	B1	
	(b) (i) ($\lambda =$) c/f or 330 / 2200 0.15 m	C1 A1	
	(ii) 1 no change and 2 increases	B1	
	(c) (i) 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules) 2 compressions and rarefactions / molecules vibrate / longitudinal wave vibration / oscillation / energy passed on	B1 B1 B1	
	(ii) fewer / no molecules / particles and less / no energy / vibration transferred	B1	[8]
6	(a) (i) X N-pole Y S-pole and Z N-pole	B1 B1	
	(ii) they touch / move towards each other and opposite poles attract	B1	
	(b) any sensible use: starting-motor circuit; with a logic gate; nuclear power station corresponding explanation: current too large for dash-board switch; current too small to power device; too dangerous to reach switch	B1 B1	[5]
7	(a) (i) supplies the (mains) e.m.f. / voltage	B1	
	(ii) to complete the circuit / is at 0 V	B1	
	(b) (i) the circuit / supply is cut / broken or current stops fuse melts / blows / burns	B1 B1	
	(ii) live wire when it cuts the circuit / melts no part of the appliance is live / no shock	B1 B1	[6]
[45]			

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Question 10 · included (a)

only $v=f\lambda$

- (a) Calculate the speed of the skier after 5.0 s.

speed =[2]

Mark scheme — 5054/22 Q10

Page 5	Mark Scheme			Syllabus	Paper
	Cambridge O Level – October/November 2016			5054	22
	(ii) 1	cylinder is magnetised (by induction) top (of cylinder) is an S-pole unlike poles attract or S-pole attracts N-pole		B1 B1 B1	
	2	it does not (remain in contact) and iron is temporary / soft magnetic material / core (and cylinder) lose magnetisation		B1	[4]
	[15]				
10	(a) (i)	$3.0 \times 10^8 \text{ m/s}$		B1	
	(ii)	$(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ $7.0 \times 10^{-7} \text{ m}$		C1 A1	[3]
	(b) (i)	decreases		B1	
	(ii)	$\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 49°		C1 A1	
	(iii)	41°		B1	[4]
	(c) (i)	dispersion at both surfaces and refractions in correct direction violet / blue light below the red light shown		B1 B1	
	(ii)	spectrum or band of (continuous) colours or colours of rainbow red, orange, yellow, green, blue, (indigo, violet)		B1 B1	
	(iii) 1	X marked above red		B1	
	2	it is / black surfaces are good absorbers (of IR radiation)		B1	[6]
	(d)	intruder / human being emits IR	IR beam broken	IR reflected	B1
		or	or		
		intruder warm or IR detected	does not reach detector	change detected	B1 [2]
	[15]				

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May/June 5054/21

Question 10 • included (b), (c)(i), (c)(iii)

not (a) mirror; not (c)(ii) hearing

- (b) Complete the equation to show the radioactive decay of $^{131}_{53}\text{I}$.



[2]

- (c) When a nucleus of $^{125}_{53}\text{I}$ decays, only a gamma ray is emitted.

Suggest why it is better for the doctor to inject the isotope $^{125}_{53}\text{I}$ into the patient.

.....

.....

.....

.....[2]

Mark scheme — 5054/21 Q10

10(b)(i) line (joining points) C1 line joining points / particles on wave with same phase or line joining points along a crest etc. A1 10(c)(i) ($v =$) $f\lambda$ or $2(000) \times 16$ C1 32 000 cm / s or 320 m / s A1 10(c)(i) ($v =$) $f\lambda$ or $2(000) \times 16$ C1 32 000 cm / s or 320 m / s A1

Question	Answer	Marks
10(a)(i)	both reflected rays correct by eye	B1
	image in correct position shown by continuation of rays behind mirror	B1
10(a)(ii)	virtual or upright or laterally inverted or same size (as object)	B1
10(b)(i)	line (joining points)	C1
	line joining points / particles on wave with same phase or line joining points along a crest etc.	A1
10(b)(ii)1	correct angle to surface $\pm 7^\circ$ with correct orientation and similar wavelength (by eye)	B1
10(b)(ii)2	at least two lines at smaller angle to surface with correct orientation in slower medium	B1
	showing smaller and constant wavelength with wavefronts deviated in correct direction	B1
10(c)(i)	$(v =) f\lambda$ or $2(000) \times 16$	C1
	32 000 cm / s or 320 m / s	A1
10(c)(ii)1	a range with 15–25 Hz as the lowest frequency and 15–30 kHz as the highest	B1
10(c)(ii)2	1.6 cm (if highest frequency is 20 000 Hz)	B1
10(c)(iii)	tube or other method to produce (narrow) beam of sound on source and/or on detector	B1
	stated detector or stated reflector	B1
	detector moved to find maximum loudness or angles i and r measured with suitable experiment	B1

May/June 5054/22

Question 6 · included (a), (b)(i)

not (b)(ii) hearing

- (a) Fig. 5.1 shows the arrangement of the molecules in the solid.

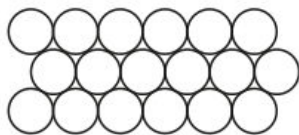


Fig. 5.1

In the space below, draw the arrangement of the molecules in the liquid.

[1]

- (b) Complete the table by describing the motion of the molecules in the solid, liquid and gas.

	motion of the molecules
solid	
liquid	
gas	

[3]

Mark scheme — 5054/22 Q6

6(a) longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions B1 transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs B1 6(b)(i) ($\lambda =$) v / f or 330 / 3800 B1 0.087 m or 8.7 cm B1

Question	Answer	Marks
6(a)	longitudinal - vibration / oscillation / movement to and fro and in direction of wave or has compressions and rarefactions	B1
	transverse – vibration / oscillation / movement up and down and at right angles to wave or has crests and troughs	B1
6(b)(i)	$(\lambda =) v / f$ or 330 / 3800	B1
	0.087 m or 8.7 cm	B1
6(b)(ii)	not heard and because below the range of audible frequencies or audible range is 20 – 20 000 Hz or too low a pitch / frequency	B1

Question	Answer	Marks
7(a)(i)	current in coil (at right angles) in a magnetic field (of magnet) or left-hand rule mentioned	B1
7(a)(ii)	reverses / changes direction of current (in coil)	B1
	reverses current every half turn / when coil is vertical or reverses forces (on side AB / CD) or keeps forces in same direction for wire on one side	B1
7(b)(i)	$(E =) VIt$ or $2 \times 12 \times 8$ or $E = Pt$ and $P = VI$ or $E = VQ$ and $Q = It$	C1
	190 or 192 J	A1
7(b)(ii)	73% or 0.73	B1

Oct/Nov 5054/22

Question 6 · included (a), (b), (c)

- (a) Calculate the pressure on the ground caused by the block.

pressure =[2]

- (b) State why the total pressure on the ground underneath the block is larger than the value obtained in (a).

.....

.....[1]

- (c) The block is of uniform density.

Mark scheme — 5054/22 Q6

6(a) reflection of sound B1 6(b)(i) decreases B1 6(c) $(\lambda =) c / f$ or 330 / 3700 C1 0.089 m or 8.9 cm or 89 mm A1 5054/22 – Mark Scheme
PUBLISHED October/November 2017 © UCLES 2017 Page 6 of 10 Question Answer Marks

Question	Answer	Marks
5(a)	joining together of (small) <u>nuclei</u> (to make bigger nuclei)	B1
	energy released	B1
	hydrogen (used) or helium (produced)	B1
5(b)	electromagnetic <u>radiation</u> / infra-red / light / ultraviolet	B1
	travels through vacuum or no medium needed	B1

Question	Answer	Marks
6(a)	reflection of <u>sound</u>	B1
6(b)(i)	decreases	B1
6(b)(ii)	does not change	B1
6(c)	$(\lambda =) c/f$ or 330/3700	C1
	0.089 m or 8.9 cm or 89 mm	A1

2018**May/June 5054/21****Question 11 · included (a), (b)****10 (a) (i)** Describe two differences between boiling and evaporation.

1.
-
-
2.
-
-

[4]

- (b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16°C

temperature of water after heating = 37°C

volume of water collected in 1.0 minute = $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = $4200 \text{ J / (kg }^{\circ}\text{C)}$

density of water = 1000 kg / m^3

- (i) Calculate the mass of water leaving the shower in 1.0s.

mass =[2]

Mark scheme — 5054/21 Q11

11(a)(i) tank with water B1 dipper (bar or point) touches or in surface of water B1 light source / stroboscope / video B1 dipper moves up and down (to create wave) B1 11(a)(ii) observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object B1 moves up and down B1 11(a)(iii) line joining points M1 that are at the same point in the motion A1 11(b)(i) difference: Q has a greater / different amplitude or phase B1 similarity: same frequency; same time for one wave (period); both transverse; oscillation up and down B1

Question	Answer	Marks
11(a)(i)	tank with water	B1
	dipper (bar or point) touches or in surface of water	B1
	light source / stroboscope / video	B1
	dipper moves up and down (to create wave)	B1
11(a)(ii)	observe (surface of) water OR place particle / cork on surface OR equivalent movable / rotating object	B1
	moves up and down	B1
11(a)(iii)	line joining points	M1
	that are at the same point in the motion	A1
11(b)(i)	<i>difference</i> : Q has a greater / different amplitude or phase	B1
	<i>similarity</i> : same frequency; same time for one wave (period); both transverse; oscillation up and down	B1
11(b)(ii)	time 300 (ms) or 0.3 (s)	C1
	($f = 1/t$) numerical or algebraic in any form	C1
	3.3 Hz	A1
11(b)(ii)	$2 (\lambda =) v/f$ or $v = \lambda f$ – numerical or algebraic in any form	C1
	0.060 m	A1

May/June 5054/22**Question 5 • included (b)***only (b)***(b)** The object has a constant speed for some of the time.**Mark scheme — 5054/22 Q5**

5(b)(i) 1st two columns correct (sound longitudinal and microwaves transverse) B1 3rd column correct microwaves electromagnetic B1

Question	Answer	Marks
5(a)	(d=) s × t in any form algebraic or numerical,	C1
	3.6×10^7 m	A1
5(b)(i)	1st two columns correct (sound longitudinal and microwaves transverse)	B1
	3rd column correct microwaves electromagnetic	B1
5(b)(ii)	Layers / molecules / particles / spring coils close together or high(er) pressure (than atmospheric)	B1

Question	Answer	Marks
6(a)	any two of - ray <u>through</u> middle of lens undeviated - ray parallel to axis passes <u>through</u> focus, 3 cm from lens - ray <u>through</u> focus on left of lens parallel to axis after lens	B2
6(b)	inverted or real	B1
6(c)	image size / object size or image distance / object distance numerical or algebraic	C1
	rays from lens intersect on diagram (for image) and 1.3–1.7	A1
6(d)	projector or photographic enlarger	B1

Oct/Nov 5054/21

Question 9 · included (a), (b)*not (c) TIR*

- (a) On Fig. 3.2, sketch the shape of the graph for loads greater than 7.0 N. [1]

- (b) All the loads are removed from the spring. A block of wood is suspended from the end of the spring. The length of the spring increases from 4.9 cm to 13.4 cm.

Determine the weight of the wooden block.

weight = [2]

Mark scheme — 5054/21 Q9

9(a) the vibration direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised B1 9(b)(i) $(f =) v / \lambda$ or $0.17 / 0.019$ C1 8.9 Hz A1

Question	Answer	Marks
8(d)(ii)	from chemical (energy) or chemical (energy) as first term	B1
	to gravitational potential (energy) or to thermal / heat (energy) as last term	B1
	to thermal / internal (energy) / heat and thermal / internal (energy) / heat	B1
8(e)	no resultant force or forces balance / cancel or tension equals weight (of bucket) or upwards force equals downward force	B1

Question	Answer	Marks
9(a)	the <u>vibration</u> direction is parallel to the wave / energy travel direction or longitudinal waves have compressions and rarefactions or transverse waves have crests and troughs or longitudinal waves cannot be polarised	B1
9(b)(i)	$(f =) v / \lambda$ or 0.17 / 0.019	C1
	8.9 Hz	A1
9(b)(ii)	(frequency) stays constant or does not change	B1
9(b)(iii)	(speed) decreases	B1
	(Fig. 9.1 shows that) wavelength decreases or refraction is towards normal or $i > r$ or top of wave (in shaded region) lags behind bottom of wave	B1
9(c)(i)	electromagnetic and transverse underlined	B1
9(c)(ii)	$n = \sin(i) / \sin(r)$ or $(r =) \sin^{-1}(\sin(i) / n)$ or $(r =) \sin^{-1}(\sin(60^\circ) / 1.6)$	C1
	33(°)	A1

Question	Answer	Marks
9(c)(iii)	1 $n = 1 / \sin(c)$ or $(c =) \sin^{-1}(1 / n)$ or $\sin^{-1}(1 / 1.6)$	C1
	39°	A1
	2 total internal reflection	B1
	$\theta > c$ or 57(°) > 39(°) or angle of incidence greater than the critical angle	B1
	3 reflected ray at P (correct angle by eye)	B1
	light in air at 30° (by eye) to vertical surface	B1

Question	Answer	Marks
10(a)(i)	same number of protons or same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^4_2\alpha$	B1
	${}^{238}_{92}\text{U}$	B1
	${}^{235}_{92}\text{U}$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ or 5 (half-lives)	C1
	relevant halving or $1 / 2^5$ or 1 / 32 or 6400 / 32	C1
	200 counts / second	A1

Question 9 · included (b), (c)

not (a) sound vs ultra

- (b) When there is an alternating current (a.c.) in the primary coil, the lamp is lit.

When there is a direct current (d.c.) in the primary coil, the lamp is **not** lit.

- (c) The half-life of cobalt-60 is 5.3 years.

- (i) State what is meant by *half-life*.

.....
.....
..... [2]

- (ii) When there is a lead sheet between the detector and the sample, the average count rate is obtained from six readings taken at one-minute intervals. The six readings are given in the table in Fig. 6.4.

reading number	1	2	3	4	5	6
count rate counts / minute	61	46	43	56	49	57

Fig. 6.4

There are reasons for suggesting that the variation in these readings is random and not because the number of cobalt-60 atoms in the sample is decreasing.

State two of these reasons.

1.
.....
2.
..... [2]

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[Turn over

Mark scheme — 5054/22 Q9

9(b)(i) transmission of energy B1 (through a medium) with no net movement of medium or by vibrating particles B1 vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised B1 9(c)(i) decreases B1

Question	Answer	Marks
8(c)	any two from: smaller time to drop to zero velocity / hit ground line not straight or velocity does not change uniformly or gradient not constant smaller area under (first part of) graph or less distance travelled slower final velocity initial downward gradient steeper	B2

Question	Answer	Marks
9(a)	frequency of sound wave small(er) or its frequency is less than 20 000 Hz	B1
9(b)(i)	transmission of energy	B1
	(through a medium) with no net movement of medium or by vibrating particles	B1
	vibrations parallel (and antiparallel) to wave / energy travel direction or cannot be polarised	B1
9(b)(ii)	1 two centres of rarefactions labelled <i>R</i>	B1
	2 distance from one point to adjacent identical point indicated (with double-headed arrow)	B1
	3. $(v =) f\lambda$ or $25\,000 \times 0.047 / 0.048 / 0.049$ or $25\,000 \times 4.7 / 4.8 / 4.9$ or $25\,000 \times 47 / 48 / 49$	C1
	1200 m / s	A1
9(c)(i)	decreases	B1
9(c)(ii)	four / five straight lines in air that touch the compressions still in the liquid and no intermediate / extra lines between the correct lines	B1
	at least four compressions in air parallel to each other	B1
	at least four straight lines at shallower angle from horizontal and slope correct	B1

Question	Answer	Marks
9(d)	object (to be cleaned) immersed in liquid / solvent	B1
	object / liquid agitated / vibrated by ultrasound	B1
	dirt (particles) shaken off or dislodges / removes dirt	B1

Question	Answer	Marks
10(a)	electrons c.a.o.	M1
	towards the ammeter or away from the negative terminal or towards the positive terminal	A1
10(b)(i)	thermistor c.a.o.	B1
10(b)(ii)	$1/R_T = 1/R_1 + 1/R_2$ or $1/R_T = 1/1.5 + 1/6.0$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	1.2 (Ω)	C1
	2.5 Ω	A1
10(b)(iii)	$(I =) V/R$ or 12 / 2.5	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current increases	B1
	voltage (across 1.3 Ω) increases	B1
	trace moves towards top of screen / upwards	B1

Question 10 • included (a)*not (b) speed of sound exp*

- (a) (i)**
- Determine the distance between the two towns.

distance = [1]

Mark scheme — 5054/21 Q10

10(a)(i) 1.7 (cm) seen or any multiple of wavelength measured on diagram C1 6.4–7.2 cm A1 10(a)(ii) number of waves/cycles/oscillations in one second / unit time C1 number of wavelengths/crests/troughs/ wave cycles/wave oscillations generated/made/pass a point in one second/unit time or number of oscillations of a source/particle in one second / unit time A1 10(a)(iii) 2.5 Hz B1 10(a)(iv) moved up and down B1 regularly or at constant frequency / 2.5 times a second B1 10(a)(v) speed smaller / decreases and wavelength smaller / decreases B1

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Question	Answer	Marks
10(a)(i)	1.7 (cm) seen or any multiple of wavelength measured on diagram	C1
	6.4–7.2 cm	A1
10(a)(ii)	number of waves/cycles/oscillations in one second / unit time	C1
	number of wavelengths/crests/troughs/ <u>wave</u> cycles/ <u>wave</u> oscillations generated/made/pass a point in one second/unit time or number of oscillations of a <u>source/particle</u> in one second / unit time	A1
10(a)(iii)	2.5 Hz	B1
10(a)(iv)	moved up and down	B1
	regularly or at constant frequency / 2.5 times a second	B1
10(a)(v)	speed smaller / decreases and wavelength smaller / decreases	B1
10(b)(i)	tape measure or trundle wheel or microphones	B1
	stopwatch or timer	B1
10(b)(ii)	start timer on seeing the smoke from gun / sound picked up by microphone	B1
	stop timer on hearing sound	B1
	measure (large) distance between students	B1
10(b)(iii)	both between and including 1000 and 10 000 m/s	B1
	solid speed > liquid speed	B1

Question 5 • included (a), (b)(i)*not (b)(ii) cleaning*

- (a)**
- The energy absorbed by solar panels comes from the Sun.

(b) Explain why solar panels are usually black.

.....

.....

.....

..... [2]

[Total: 5]

Mark scheme — 5054/22 Q5

5(a) vibration / oscillation of particles or energy transfer or compressions and rarefactions B1 vibration / oscillation (of particles) parallel to energy travel direction / propagation direction B1 5054/22 – Mark Scheme PUBLISHED October/November 2019 © UCLES 2019 Page 6 of 10 Question Answer Marks 5(b)(i) ($\lambda =$) v / f or $1500 / 42\,000$ C1 0.036 m or 3.6 cm A1

Question	Answer	Marks
3(a)(i)	(nuclear) <u>fusion</u> (reactions) or nuclei fuse	B1
	small <u>nuclei</u> / hydrogen <u>nuclei</u> joining together (and release energy) or larger <u>nucleus</u> produced	B1
3(a)(ii)	infra-red (radiation) or ultraviolet (radiation) or (visible) light	B1
3(b)	black surfaces are (good) absorbers	M1
	of (infra-red) radiation / heat / (visible) light / thermal energy	A1

Question	Answer	Marks
4(a)	any two from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape or slow(est) molecules / molecules with small(est kinetic) energy remain or (average) speed of remaining molecules decreases or molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature or takes too much time	B1

Question	Answer	Marks
5(a)	vibration / oscillation of <u>particles</u> or energy transfer or compressions and rarefactions	B1
	vibration / oscillation (of particles) parallel to energy travel direction / propagation direction	B1

Question	Answer	Marks
5(b)(i)	$(\lambda =) v / f$ or 1500 / 42 000	C1
	0.036 m or 3.6 cm	A1
5(b)(ii)	(ultrasound) vibrates (the cleaning fluid / jewellery)	B1
	dirt shaken off	B1

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	$(V =) 12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	$(R =) V / I$ or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

Question	Answer	Marks
7(a)(i)	current and <u>magnetic field</u> / <u>magnetic flux</u> (experiences a force) or the two <u>magnetic fields</u> interact	B1
7(a)(ii)	three separate approaches I / II / III	
7(a)(iii)	B to A	B1
	left-hand rule mentioned / described	B1
	or	

Question 4 • included (a)*not (b) critical angle/TIR*

- (a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge.
The force P is 25 N.

Mark scheme — 5054/21 Q4

4(a)(i) it decreases B1 4(a)(ii) it does not change B1

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Question	Answer	Marks
3(a)(i)	($I = $) $Fx_{\perp}$ or 25×0.72	C1
	18 N m	A1
3(a)(ii)	(WD =) $Fx_{\parallel}$ or $F\pi r/2$ or $25 \times 4.5/4$ or 25×1.13	B1
	28 J	B1
3(b)	moment due to force P is greater	B1
	distance of X to hinge is greater than distance from where force F acts	B1

Question	Answer	Marks
4(a)(i)	it decreases	B1
4(a)(ii)	it does not change	B1
4(b)(i)	($c = $) $\sin^{-1}(1/n)$ or $\sin^{-1}(1/1.6)$ or $\sin^{-1}(0.625)$	C1
	39°	A1
4(b)(ii)	reflected ray at Z and $i = r$ and no refracted ray	B1
	ray strikes vertical face perpendicularly and undeviated in air	B1

Question	Answer	Marks
5(a)	X, Y and Z to be two (or three) different metals	B1
5(b)(i)	(output / thermoelectric) electromotive force / e.m.f. / voltage	B1
5(b)(ii)	linearity indicates whether the output (voltage) is directly proportional to something	B1
	directly proportional to the temperature difference (between the junctions)	B1

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2021**May/June 5054/21****Question 4 • included (a)***not (b) other-topic corrections*

(a) The results are shown in Fig. 2.2.

Draw a line linking each type of surface with the appropriate meter reading. One line has been drawn for you.

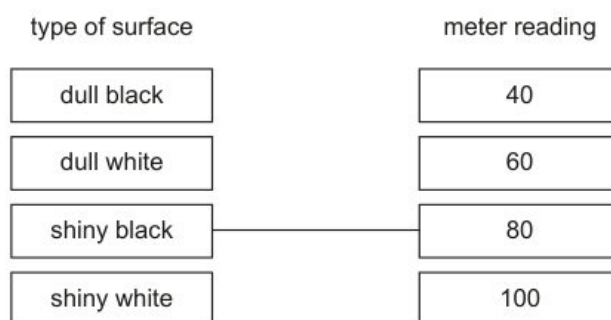


Fig. 2.2

[2]

Mark scheme — 5054/21 Q4

4(a)(i) Fig.4.1 transverse and Fig.4.2 longitudinal B1 4(a)(ii) wave drawn with larger wavelength B1 4(a)(iii) backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement) B1

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Question	Answer	Marks
3(a)	$(E =) mc^2$ in any form algebraic or numerical	C1
	$2 \times 460 \times 450$	C1
	$4.1 \times 10^5 \text{ J}$	A1
3(b)(i)	any one of: - boiling occurs at one temperature / boiling point or evaporation occurs at any temperature / below boiling point - boiling occurs within liquid or evaporation occurs on the surface - evaporation increased by draughts, more surface area (and boiling does not)	B1
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) or (molecules) do not have the energy to escape / break bonds	B1
	fewer fast / energetic molecules escape or (only) fastest / high KE molecules escape	B1
Question	Answer	Marks
4(a)(i)	Fig.4.1 transverse and Fig.4.2 longitudinal	B1
4(a)(ii)	wave drawn with larger wavelength	B1
4(a)(iii)	backwards and forwards, left and right, to and fro, vibration parallel to arrow / direction of wave (movement)	B1
4(b)	frequency	B1
	ultrasound	B1
	gamma	B1

May/June 5054/22

Question 10 • included (a)

not (b) spectrum

(a) Describe how the reading on the newton meter is used to find the mass of the brick.

.....
 [1]

Mark scheme — 5054/22 Q10

10(a)(i) 3 (.0) cm B1 10(a)(ii) number of waves per second or $5 / 10$ or $f = 1 / T$ C1 2(.0) Hz A1 10(a)(iii) $(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form C1 6(.0) cm / s A1 10(a)(iv) frequency stays the same and speed increases B1 5054/22 – Mark Scheme PUBLISHED May/June 2021 © UCLES 2021 Page 15 of 16 Question Answer Marks 10(a)(v) ■ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in- between B1 constant wavelength shown between all refracted wavefronts B1 all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal B1

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Question	Answer	Marks
9(c)(i)	$(E =) mcT$ numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) 460 + 5200 or 5660 (J) seen or 5200 + (energy in 9(c)(i) + 5200)	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

Question	Answer	Marks
10(a)(i)	3 (.0) cm	B1
10(a)(ii)	number of waves per second or $5 / 10$ or $f = 1 / T$	C1
	2(.0) Hz	A1
10(a)(iii)	$(v =) f \lambda$ or $s = d / t$ numerical or algebraic in any form	C1
	6(.0) cm / s	A1
10(a)(iv)	frequency stays the same and speed increases	B1

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Question	Answer	Marks
10(a)(v)	≥ three consecutive incident wavefronts meeting straight refracted wavefronts on the boundary without any wavefronts in-between	B1
	constant wavelength shown between all refracted wavefronts	B1
	all wavefronts refracted in correct direction, i.e. lines between the vertical and the normal	B1
10(b)(i)	four colours from violet, indigo, blue, green, yellow, orange, red	B1
	correct order of four of the above colours	B1
10(b)(ii)	prism	B1
	slit (to produce narrow beam)	B1
	dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face	B1
	correct refraction on first and second faces for all rays shown	B1

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case or prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out or <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected or <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1

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Question 8 · included (b)(iii), (b)(v)

only long/trans and $v=f\lambda$

- (b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

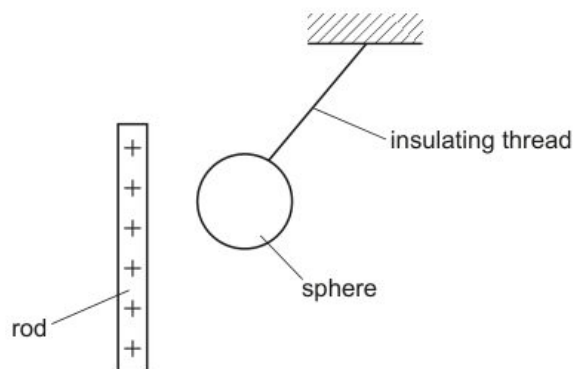


Fig. 5.1

- (i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]
- (ii) Explain why the sphere is pulled towards the rod.

.....

.....

.....

..... [2]

[Total: 6]

Mark scheme — 5054/21 Q8

8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 © UCLES 2021 Page 9 of 11 Question Answer Marks 8(b)(i) the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses B1 the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses B1 5054/21 – Mark Scheme PUBLISHED October/November 2021 © UCLES 2021 Page 9 of 11 Question Answer Marks

Question	Answer	Marks
7E(a)	magnetic field lines cut the solenoid / coils or changing magnetic field in solenoid	B1
	an electromotive force / e.m.f. is <u>induced</u>	B1
	electromotive force / e.m.f. produces the current	B1
7E(b)	negative / opposite / reversed current / deflection / reading	B1
	negative e.m.f. / voltage (induced) or magnetic field cut in opposite direction or magnetic field changes in opposite sense / decreases (owtte)	B1
7O(a)	NOR (gate)	B1
7O(b)(i)	lower logic gate has an input of logic level 1 (and so its output / Y is 0) or the output X has a logic level 1 and so the inputs must both be 0 (Y is an input to the upper NOR gate)	B1
7O(b)(ii)	X changes to 0 and Y changes to 1.	B1
7O(b)(iii)	both X and Y stay at the new values or neither X nor Y changes	B1
7O(b)(iv)	the output shows which terminal (P or Q) last had the value 1.	B1

Question	Answer	Marks
8(a)(i)	magnetic field or magnet	B1
8(b)(i)	the current (in coil continually) reverses (direction) or the force (on the coil continually) reverses	B1
	the current (in the coil continually) reverses (direction) and the force (on the coil continually) reverses	B1

Question	Answer	Marks
8(b)(ii)	cone moves forwards and air is compressed / hits (air) molecules	B1
	cone moves backwards and rarefaction produced / molecules sucked back	B1
	(air) molecules vibrate or production of compressions and rarefactions continues	B1
8(b)(iii)	vibration direction is parallel to energy travel / propagation direction	B1
	in <u>transverse</u> wave, vibration direction is perpendicular to propagation / energy travel direction	B1
8(b)(iv)	(time period =) 0.0050 (s) or 1 / 0.0050 or 200 (times per second)	C1
	400 (times per second)	A1
8(b)(v)	$(\lambda =) c / v$ or 340 / 200	C1
	1.7 m	A1
8(c)(i)	decreases / less loud / quieter	B1
	amplitude of sound / vibration decreases or (maximum) force on coil decreases	B1
8(c)(ii)	(pitch) stays the same / unchanged and no change to frequency / time period	B1

Question	Answer	Marks
9(a)(i)	molecules with similar / slightly smaller packing density and most molecules touching	B1
	random arrangement	B1
9(a)(ii)	molecules (of gas) further apart	B1
	(repulsive) forces between molecules (much) smaller	B1

Question 3 • included (a)

not (b) TIR

(a) State what is meant by a *vector*.

.....
..... [1]

Mark scheme — 5054/22 Q3

3(a)(i) it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction B1 change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first B1
3(a)(ii) (wavelength) decreases B1 (frequency) does not change. B1

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Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force or no resultant force or forces balance	B1

Question	Answer	Marks
3(a)(i)	it / light refracts / bends / moves towards normal or angle of incidence greater than angle of refraction	B1
	change in direction of the ray / refraction shows a change in speed (at the boundary) or $\sin(i) / \sin(r)$ is ratio of speeds or top of ray slows down first and enters the block first	B1
3(a)(ii)	(wavelength) decreases	B1
	(frequency) does not change.	B1
3(b)(i)	$(\sin(r) =) \sin(i) / n$ or $\sin(45^\circ) / 1.6$ or $0.71 / 1.6$ or 0.44 or $(r =) 21^\circ$	C1
	26°	C1
	64°	A1
3(b)(ii)	64° / angle of incidence / i is greater than the critical angle / 39° or angle of incidence = 64°	B1
	it undergoes total internal reflection	B1

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1

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Question 8 • included (a), (b), (c)

not (d) satellite

- 6 (a) Fig. 6.1 shows part of a toy which contains two ring-shaped, permanent magnets. A plastic rod passes through the centre of both magnets.

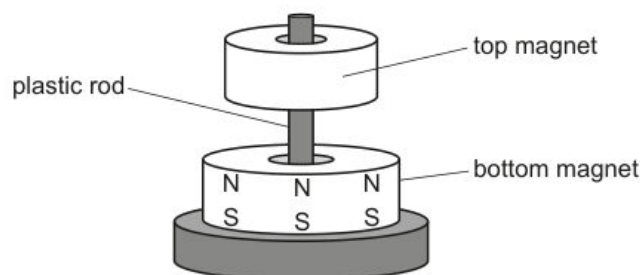


Fig. 6.1

The top magnet can move up and down freely around the plastic rod.

The magnetic poles on the bottom magnet are shown in Fig. 6.1.

- (b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0 V, as shown in Fig. 5.2.

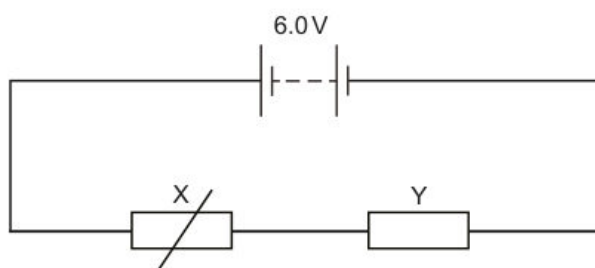


Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is $600\ \Omega$ and the current in thermistor X is 0.0020 A.

Mark scheme — 5054/22 Q8

8(a)(i) 3 B1 8(a)(ii) shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change) B1 1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i) B1 8(b)(i) move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight B1 8(c)(i) (piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary B1

Question	Answer	Marks
8(a)(i)	3	B1
8(a)(ii)	shape approx. sinusoidal with amplitude 3 mm for the first complete wave (may then change)	B1
	1 wave taking 0.5 s for first wave shown within half a square or 1.5 s shown for the number of waves in (a)(i)	B1
8(b)(i)	move bar up and down more often (per second) / faster or make rubber band(s) strong(er) / short(er) / tight(er) / more in number or use bar of less mass / weight	B1
8(b)(ii)	speed remains the same	B1
	wavelength decreases	B1
8(c)(i)	(piece of) plastic / (piece of) glass / block / plate or a shallow(er) / deep(er) area / boundary	B1
8(c)(ii)	diagram showing within the ripple tank:	B1
	crests parallel to bar hitting boundary with a boundary labelled	
	- correct refraction of plane crests either towards or away from normal on the same side of the normal - or no direction change if crests are initially parallel to boundary but with a wavelength change	B1
	different wavelength shown before and after refraction and constant before and after (by eye)	B1
8(d)(i)	($t =$) distance / speed algebraic or numerical seen in any form	C1
	$2 \times 560(000)$ seen or used or 0.0019 (s)	C1
	0.0037 s	A1

Question	Answer	Marks
8(d)(ii)	One of (both): - reflect / refract - have same speed (in a vacuum) - travel in a vacuum / don't need a medium - are electromagnetic - are transverse - carry energy / cause heating - have larger wavelength / smaller frequency than visible light	B1
8(d)(iii)	less likely to be hacked / intercepted / greater security or more channels / better bandwidth possible / more data or less likely for transmission to be interrupted or cost of satellites saved / satellites expensive	B1

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) - two electrons (and ion has 1 electron) - one more electron (than the ion) - no charge (and ion positive) - same number of electrons as protons;	B1
9(a)(ii)	alpha particle has (any 1 from) : - no electrons - charge +2 (e) - only protons and neutrons or He ion has an electron	B1

Question 4 • included (b)(i)

only frequency definition

- (i) The height of the trapped air in the tube is x .

Explain, in terms of molecules, why x changes when the air is heated.

.....

.....

.....

.....

..... [3]

Mark scheme — 5054/21 Q4

4(b)(i) number of oscillations / cycles / wavelengths B1 per unit time B1

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Question	Answer	Marks
4(a)	electromagnetic (waves) or components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage or X-rays show the bone structure rather than foetal development or image would not be clear	B1

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Question 4 • included (b), (c)

not (a) vacuum exp; not (d) hearing

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N/kg.

(c) Describe what is meant by 'a rarefaction'.

.....
 [1]

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Mark scheme — 5054/22 Q4

4(b) (they) vibrate M1 (vibration) parallel to / in the direction of propagation / of energy travel A1 4(c) (point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average) B1

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Question	Answer	Marks
3(a)	chemical potential energy and no other energy	B1
3(b)(i)	radiation or infrared mentioned	B1
	radiation and infrared (travel to the man)	B1
	energy / infrared / radiation <u>absorbed</u> by the man / pullover or travels in straight line or travels at the speed of light or increases (man's) internal energy	B1
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	B1
	less (thermal) energy escapes or more (thermal) energy reaches / absorbed by man	B1
3(b)(iii)	EITHER black (surface) is good absorber / poor reflector of infrared radiation	B1
	more (thermal) energy absorbed or more thermal energy transferred to man	B1
	OR trapped air and reduced convection / air poor conductor / air is a good insulator	(B1)
	less (thermal) energy lost	(B1)

Question	Answer	Marks
4(a)	equipment e.g. (evacuable) container and (suspended electric) bell / source of sound and (air) <u>pump</u>	B1
	action e.g. ringing / sound (produced) / sound heard and evacuate air / vacuum (produced)	B1
	observation e.g. transmission / hearing of sound ceases / no sound	B1

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Question	Answer	Marks
4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1
4(d)	$(\lambda =) v / f$ or 1500 / 140 000	C1
	1.1×10^{-2} m or 1.1 cm or 11 mm	A1

Question	Answer	Marks
5(a)	energy / work done per unit charge or $\frac{\text{energy}}{\text{charge}}$ or $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply or energy used driving charge around a (complete) circuit or energy transferred to electrical energy	B1
5(b)	it lasts for a longer time or more (electrical) energy available or power supply still functions if one cell fails / runs flat or can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases c.a.o.	B1
	current change or voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases c.a.o.	B1
5(c)(ii)	$(V_1 =) V_0 \times R_1 / R_0$ or $1.5 \times 4000 / (4000 + 8000)$ or $1.5 \times 4000 / 12\ 000$ or $I = V / R$ or $1.5 \times 8000 / (4000 + 8000)$ or $I = 1.5 / 12\ 000$ or 0.125 mA	C1
	0.50 V	A1

2023**May/June 5054/21****Question 7 · included (a), (b), (c)****(a)** Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

- (b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = kg [3]

- (c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = m [2]

Mark scheme — 5054/21 Q7

7(a) wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests B1
7(b) at least 3 semicircles centered on the gap B1 wavelength constant and the same on right and left of gap and some bending B1
7(c)(i) distance between two (consecutive) identical points such as two (consecutive) crests B1

Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz	A1
7(c)(iii)	1.3 (cm)	B1

May/June 5054/22**Question 5 · included (a), (b)**

- (a) (i) Describe why rotating the tube changes the pressure of the gas in the sealed end.

.....

 [1]

- (b) In Fig. 3.1 the length of the mercury column is 0.30 m.

The density of mercury is $14\,000\text{ kg/m}^3$.

Atmospheric pressure is $1.0 \times 10^5\text{ Pa}$.

Calculate the pressure of the gas in the tube.

pressure = Pa [3]

Mark scheme — 5054/22 Q5

5(a) seismic S-waves and X-rays B1 5(b)(i) (wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$ C1 frequency 6.7 (Hz) A1
 wavelength 4.0 (cm) A1

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	vibrate / oscillate with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

Question	Answer	Marks
5(b)(iv)	less diffraction or less bending	B1

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A)	B1
	voltmeter V_2 9(0 V)	B1
6(a)(ii)	($R =$) V / I in any form	C1
	36 (Ω)	A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference	B1
	provided the temperature is constant	B1
6(b)	0.35 (A) seen	B1
	or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$	
	voltage (across R) 0.35×36 or 12.6 (V) seen	B1
	19 V	B1

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1

Question 4 • included (a), (b)(i)

not n or ray path

(a) Define what is meant by 'acceleration'.

.....

.....

..... [2]

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....

..... [1]

Mark scheme — 5054/21 Q4

4(a) (■ =) v/f or $3.0 \times 10^8 / 4.7 \times 10^{14}$ ■ $108 / 4.7$ ■ 1014 C1 6.4×10^6 (m) or $3 / 4.7 \times 10^6$ ■ 106 C1 6.4×10^{-7} (m) A1 4(b)(i) speed: decreases frequency: stays the same wavelength: decreases B2

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Question	Answer	Marks
4(a)	$(\lambda =) v/f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^6 (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

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Question 5 • included (b)

only trans vs long

(b) The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17°C and the heater is switched on.

Mark scheme — 5054/21 Q5

5(b) in a transverse wave, the vibration direction is perpendicular / at right angles to the propagation direction B1 in a longitudinal wave, the vibration direction is parallel to the propagation direction B1 Question Answer Marks

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Question	Answer	Marks
4(a)	$(\lambda =) v/f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1
	emergent ray parallel to initial incident ray	B1

Question	Answer	Marks
5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

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Question	Answer	Marks
5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	$(R =) V/I$ or $1.5 / 0.20$ or $7.5 (\Omega)$	C1
	$(R_T =) 7.5 - 1.5 - 3.5$	C1
	$2.5 (\Omega)$	A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases or smaller fraction of total resistance	B1
	p.d. across R and S increases or (p.d. is) smaller fraction of e.m.f.	B1

Question	Answer	Marks
7(a)(i)	P: (carbon) brush and Q: slip ring(s)	B1
7(a)(ii)	to connect the output terminal(s) and the (rotating) coil/brushes	B1
	without wires twisting	B1
7(b)(i)	at least one of the five positions when the output e.m.f. is zero indicated with X and no indications elsewhere	B1

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Question 10 · included (b)

only $v=f\lambda$

(b) The weight produces a moment about the pivot through H.

Mark scheme — 5054/22 Q10

10(b) (■ =) $v/f = 3.0$ ■ 108 / 8.8 ■ 1019 C1 3.4 ■ 10–12 (m) A1

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Question	Answer	Marks
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	B1
	outward force (due to high temperature / nuclear reaction) or force due to high temperature / nuclear reaction or radiation pressure	B1
	balances (gravitational) force (inwards)	B1
9(c)	(red supergiant explodes as) supernova (explosion)	B1

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ and no other path indicated as γ	B1
10(a)(ii)	beta (β -radiation)	B1
	left-hand rule (mentioned)	B1
	first finger into page and thumb downwards	B1
	current opposite to motion or particles negative (and so are beta)	B1
10(a)(iii)	beta (β -radiation) completely absorbed	B1
	<u>some</u> gamma rays absorbed or (some) gamma passes through	B1
10(b)	$(\lambda =) v/f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	3.4×10^{-12} (m)	A1

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Question 4 · included (a)(i), (a)(ii), (b), (c)

not (a)(iii) ultrasound use

(a) (i) On Fig. 1.2, mark:

- with the letter P **one** point where the ball accelerates
- with the letter Q **one** point where the ball has constant speed.

[2]

- (b) The speeds of the ball at A, B, C and D are v_A , v_B , v_C and v_D respectively.

Arrange these four speeds from slowest to fastest.

slowest $\longrightarrow$ fastest

--	--	--	--

[1]

- (c) Work is done to transfer energy between energy stores as the ball moves from A to B.

Name the force involved in the work done and describe the energy transfer.

.....

.....

.....

..... [3]

Mark scheme — 5054/21 Q4

4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) ($V =$) $f\lambda$ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(a)(i) number of (complete) waves (passing a point) per second B1 4(a)(ii) ($V =$) $f\lambda$ in any form C1 330 m / s A1 4(a)(iii) ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance B1 4(b)(i) longitudinal wave - oscillation backwards and forwards / in direction of travel of wave B1 transverse wave - oscillation at right angles to (direction of travel of) wave B1 diagram of each type B1 4(c) long wavelengths bend around hill and short wavelengths do not bend B1 (due to) diffraction B1 Question Answer Marks

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Question	Answer	Marks
3(b)(v)	(liquid) water has a higher specific heat capacity; water has stronger bonds	B1

Question	Answer	Marks
4(a)(i)	number of (complete) waves (passing a point) per second	B1
4(a)(ii)	($V =$) $f\lambda$ in any form	C1
	330 m / s	A1
4(a)(iii)	ANY one from cleaning, prenatal and other medical scanning, sonar, calculation of depth or distance	B1
4(b)(i)	longitudinal wave - oscillation backwards and forwards / in direction of travel of wave	B1
	transverse wave - oscillation at right angles to (direction of travel of) wave	B1
	diagram of each type	B1
4(b)(ii)	earthquake (waves) or other correct source which is a single source that generates both types of wave, e.g. lightning	B1
4(c)	long wavelengths bend around hill and short wavelengths do not bend	B1
	(due to) diffraction	B1

Question	Answer	Marks
5(a)	correct symbol; 	B1
5(b)(i)	correct curvature	B1
5(b)(ii)	(at higher current / voltage) temperature increases	B1
	(at higher current / voltage) resistance increases	B1

Question 5 · included (a), (b), (c)

- (a) Explain, in terms of electrons and the atoms of the metal, how thermal energy is transferred through the walls of the metal cylinder.

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.....

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.....

.....

..... [3]

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E IN THIS MA

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

- (b) The piston is fixed in position so the pressure of the gas increases as the temperature of the gas increases.

DO NOT WRITE IN THIS MARGIN

- (c) The gas in the cylinder reaches a temperature of 100°C .

The piston is released and moves to the right. The temperature remains at 100°C .

The pressure of the gas now decreases.

- (i) Explain, in terms of the particles of the gas, why the pressure of the gas now decreases.

.....

 [1]

- (ii) Eventually, the piston stops moving.

Explain why.

.....

 [2]

[Total: 10]



Mark scheme — 5054/21 Q5

5(a) ultrasound / sound / it is transmitted by vibrating matter / particles B1 no matter / particles in a vacuum / no medium B1 5(b) (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction B1 in a transverse wave, vibration (of particles) is perpendicular to the propagation direction B1 5(c)(i) (■ =) v / f or $1500 / 8.4$ ■ 106 C1 1.8 ■ 10N C1 1.8 ■ 10^{-4} (m) A1

Question	Answer	Marks
4(c)(ii)	frequency increases and wavelength decreases	B1
4(d)	infrared (radiation)	B1

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	B1
	no matter / particles in a vacuum / no medium	B1
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	$(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$	C1
	1.8×10^N	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

2025**May/June 5054/21****Question 5 · included (a), (b), (c)****(a)** Describe the transfers between energy stores that occur as the load is lifted.

.....

.....

.....

..... [3]

(b) The efficiency of the motor, pulley and load system is less than 100%.

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(c) A tall bus is tested for stability.

Fig. 2.2 shows the bus on a slope.

The centre of gravity of the bus is marked.

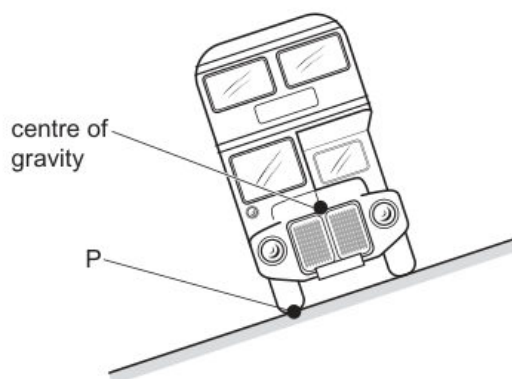


Fig. 2.2

Mark scheme — 5054/21 Q5

5(a)(i) C and R marked correctly B1 5(a)(ii) compressions are where pressure is high / particles close together / density is high B1 5(b)(i) 0.75 (Hz) B1 5(c) spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole B1 5054/21 – Mark Scheme PUBLISHED May/June 2025 © Cambridge University Press & Assessment 2025 Page 12 of 15 Question Answer Marks

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Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	<i>speed</i> : no change	B1
	<i>wavelength</i> : decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

May/June 5054/22

Question 4 • included (a), (b), (c)

(a) Define 'power'.

.....
 [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power =W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

Mark scheme — 5054/22 Q4

4(a) motion of particles: backwards and forwards / to and fro / closer and further apart B1 (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions B1

4(b)(i) number of oscillations per second or number of wave(length)s (passing a point) per second B1 4(c)(i) (at least) three correctly curved wavefronts centered on gap B1 same wavelength as on left and some curving in correct direction B1

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Question	Answer	Marks
4(a)	<i>motion of particles:</i> backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions	B1
4(b)(i)	number of oscillations per second or number of wave(length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f = 1 / T$ or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown and at least one complete oscillation seen	B1
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left and some curving in correct direction	B1
4(c)(ii)	diffraction	B1

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Question 5 • included (b)(i)

only $v = f\lambda$

- (i) Calculate the energy transferred from the gravitational potential energy store of the container.

energy transferred = J [2]

Mark scheme — 5054/22 Q5

5(b)(i) (■ =) c f or 8 3.0 10 1.2 10 N ■ ■ C1 2.5 ■ 10X (m) C1 0.025 (m) A1

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Question	Answer	Marks
5(a)	UV M → increasing wavelength	B1
5(b)(i)	$(\lambda =) \frac{c}{f} \text{ or } \frac{3.0 \times 10^8}{1.2 \times 10^N}$	C1
	$2.5 \times 10^X \text{ (m)}$	C1
	0.025 (m)	A1
5(b)(ii)	encoded microwave / microwave signal / microwave carrying data / microwave transmitting data (to satellite from Earth)	B1
	satellite transmits microwave to Earth / dish / antenna / subscriber / television	B1
5(c)	to prevent microwaves escaping / leaving the oven / spreading out of the device	B1
	microwaves / they can cause burns (in living tissue / people) / damage to tissue through heating / blisters	B1

Question	Answer	Marks
6(a)	a d.c. / it has one direction or does not (repeatedly) reverse direction or polarity / direction does not change	B1
6(b)(i)	$(Q =) It \text{ or } 41 \times 120$	C1
	$4.9 \times 10^3 \text{ (C)}$	A1